

Predicting the Zero-Shot Transfer of a Promptable Video Tracker

Weekly update — stress-testing the negative result

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This week, in one sentence

I tried hard to **break** last week's "we can't predict it" result — on **two new datasets** and by **asking SAM 3 itself** — and it **held everywhere**.

- The “no” is now **triangulated**: same answer on a camera-trap dataset, an internet-video dataset, and a model-internal check.
- That turns a one-dataset result into a **robust, honest negative** — the contribution.
- Everything below is built, hermetically tested, run on the GPU, committed, and written up.

Where we were last week

On **SA-FARI** we can *measure* SAM 3's reliability but **cannot predict it in advance** from label-free “distance” signals — beyond a small, obvious **animal-size** effect.

- Clean negative result, defended by a positive control (the same test *does* catch size).
- **The gap:** it was one model, one dataset, one inference run — so not truly *independent* evidence, and it left one before-running idea untried.
- **This week closes both.**

What I did this week

1. ✓ **Re-ran the whole pipeline on two independent datasets** — via offline adapters, so nothing in the pipeline changed; a new dataset is just converted to the format the loader already reads.
 - **MammAlps** — Alpine camera traps, 5 species, box-only ground truth.
 - **BURST** — internet/movie video, the 41-species animal subset of a mask-native benchmark.
2. ✓ **Implemented the last before-running idea** — read familiarity from **SAM 3's own feature space** (hedging the fact that its true training data is secret).
3. ✓ **Wrote it all into the dissertation** — methodology, results, discussion, tables, and a plain-English explainer for each; fixed bibliography + layout.

Engineering: box→mask adapters, WordNet animal filter, HTTP-range frame streaming (no huge downloads), SAM 3 encoder embeddings — all hermetically tested before the GPU runs.

Replication 1 — MammAlps (small, camera-trap)

- Different continent, species, and camera rig; box-only, so I score box-overlap. **20 report cards, 5 species.**
- **Result: the distances predict nothing** — same as SA-FARI.
- But it is a **consistency check, not proof**: at 20 cards the test is **underpowered** — even a feature I *know* is predictive can't pass it here.

The one “significant” feature was a fake

Visual distance cleared the bar — but pointing the **opposite way** to the theory (more-unusual animals detected *better*) and driven by animal size. The **same confound** SA-FARI showed, not a finding.

Replication 2 — BURST (bigger, mask-native, many species)

- **41 species, 132 report cards**, real masks, genuine “following” content — the powered test MammAlps couldn’t be. SAM3 works here (score ≈ 0.64).
- **Result: a third convergent null** — the distances still predict nothing.
- Far better powered than the tiny MammAlps by sheer numbers — but a **convergent** null, not proof: no before-running signal fires here (not even animal size), so a small effect can’t be ruled out.

Three datasets now agree. The “no” is not a quirk of camera-trap data or of one inference run.

The picture across all three datasets

Predict SAM 3's score from...	SA-FARI	MammAlps	BURST
the four before-running distances	— no	— no	— no
animal size (the confound)	✓ a little	— no	— no

- No *before-running* label-free distance predicts transfer, anywhere.
- The *only* thing that ever reaches significance is the **animal-size** confound — modestly, and only on SA-FARI — which doubles as the **positive control**: the same test that stays flat for the distances *does* catch size out-of-sample, so the “no” is real, not a weak test.

The last idea: ask SAM 3 how well it “knows” the animal

The hole it plugs

“Novelty” has always meant distance from *our* pile — but **nobody knows what SAM 3 was actually trained on**. So maybe our distances point at the wrong reference.

- **The trick:** instead of a pile we picked, ask the model. Push each animal through **SAM 3’s own eyes** and see how distinctly it sits in its feature space.
- Familiar animal → a clean, distinct blob; genuinely new → fuzzy, smeared into others.
- Still **before-running** and **label-free** (a property of the picture, not of SAM 3’s output).

...and the answer is still no

- Tried three ways to measure “distinctness”; ran it on **SA-FARI and BURST**.
- **None predicts transfer** at the strict bar. And all three turn out to be **63–85% the same number as the old visual distance** — they just re-measure “how different does it look,” already a dead end (and a size confound).
- On the powered SA-FARI cells the animal-size control still validates — so this is a **genuine no**, not a weak test.

Why it matters anyway

It **closes the loophole**: even reading familiarity *from the model itself* — sidestepping the “you don’t know its training data” objection — you *still* can’t forecast its transfer.

Where this leaves us

The negative result is now **robust and triangulated** — one honest “no,” the same across three datasets and a model-internal check.

- Powered enough (SA-FARI, BURST) to rule out a moderate effect; the small-sample caveats are stated honestly, not hidden.
- All built, GPU-run, committed, and written into the dissertation (chapters, tables, figures, PDFs).
- Optional next steps — none of which change the answer — a larger mask-native dataset for even more power, or a fuller probe of what SAM 3's representations encode.